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BS CIS Program Information

The Bachelor of Science with a major in Computer and Information Science (BS CIS) offers students a broad, liberal education in the physical sciences, mathematics, and humanities, along with intensive study in computer science.

>> CIS Course Requirements

CIS Core



All BS CIS students must take the set of eight courses outlined below:

Core CSE Cores

Course	Title	Credits	Prerequisites and Notes
CSE 2221	Software I: Software Components	4	Must have credit for CSE 1222, CSE 1223, CSE 1224, or ENGR 1281, or have passed the 2221 placement test
CSE 2231	Software II: Software Development and Design	4	Must have credit for CSE 2221 and Math 1151
CSE 2321	Foundations I: Discrete Structures	3	Must have credit for CSE 2221 and Math 1151
CSE 2331	Foundations II: Data Structures and Algorithms	3	Must have credit for CSE 2231 and CSE 2321, and STAT 3470

Course	Title	Credits	Prerequisites and Notes
CSE 2421	Systems I: Introduction to Low-Level Programming	4	Must have credit for CSE 2231 and CSE 2321
CSE 2431	Systems II: Introduction to Operating Systems	3	Must have credit for CSE 2421
CSE 3341	Principles of Programming Languages	3	Must have credit for CSE 2331, CSE 2421, and CSE 3901, CSE 3902 or CSE 3903
CSE 2501 or Philos 2338*	Social, Ethical, and Professional Issues in Computing or Ethics in the Professions: Introduction to Computing Ethics and Effective Presentation	1	For CSE 2501, must have credit for CSE 2231, CSE 2321, and GE foundation writing and info literacy course

*For students who entered the program in or after AU22. Students who entered the program before AU22 or have permission to complete the Legacy Gen Ed may take Philos 1338.

CIS Core Choices



In addition to the core CSE courses, which are designed to provide a breadth of knowledge, BS CIS students choose five additional courses. Students have the option to choose these courses independently or use these electives to fulfill some of their specialization requirements. The decisions BS CIS students must make are as follows:

Software Engineering or Databases

Course	Title	Credits	Prerequisites and Notes	Specialization Requirement
CSE 3231	Software Engineering Techniques	3	Must have credit for CSE 3901, CSE 3902, or CSE 3903. Not open to students with credit for 5231	Software Engineering
CSE 3241	Introduction to Database Systems	3	Must have credit for CSE 2123, CSE 2231, CSE 2321, or CSE 2111 and Math 2366. Not open to students with credit for 5241	Database Systems and Data Analytics

Computer Architecture or Networking

Course	Title	Credits	Prerequisites and Notes	Specialization Requirement
CSE 3421	Introduction to Computer Architecture	3	Must have credit for CSE 2231 and CSE 2421 or ECE 2560 and ECE 2060. Not open to students with credit for 5421	Computer Systems
CSE 3461	Computer Networking and Internet Technologies	3	Must have credit for CSE 2421, CSE 3430 or CSE 2451. Also must have completed ECE 2560. Not open to students with credit for 5461	Information and Computation Assurance, Computer Networking, Computer Systems (elective)

AI or Game and Animation

Course	Title	Credits	Prerequisites and Notes	Specialization Requirement
CSE 3521	Survey of Artificial Intelligence I: Basic Techniques	3	Must have credit for CSE 2331 or CSE 5331, Math 2174 or 2568 or 4568 or 5520H, and Stat 3201 or 3450 or 3460 or 3470 or 4201 or Math 4530 or 5530H. Not open to students with credit for 5521.	Artificial Intelligence
CSE 3541	Computer Game and Animation Technique	3	Must have credit for CSE 3901, 3902 or 3903. Not open to students with credit for 5541.	Computer Graphics and Game Design

Project

Course	Title	Credits	Prerequisites and Notes	Specialization Requirement
CSE 3901	Project: Design, Development, and Documentation of Web Applications	4	Must have credit for CSE 2231; and 2321; and 2421 or 3430, or 2451 and ECE 2560	

Course	Title	Credits	Prerequisites and Notes	Specialization Requirement
CSE 3902	Project: Design, Development, and Documentation of Interactive Systems	4	Must have credit for CSE 2231; and 2321; and 2421 or 3430, or 2451 and ECE 2560	Computer Graphics and Game Design
CSE 3903	Project: Design, Development, and Documentation of System Software	4	Must have credit for CSE 2231; and 2321; and 2421 or 3430, or 2451 and ECE 2560	

Capstone

Course	Title	Credits	Prerequisites and Notes	Specialization Requirement
CSE 5911	Capstone Design: Software Applications	4	Must have credit for CSE 3231, or 5231; and 2501, or Philos 1338, or 2338; and CSE 3901, 3902, or 3903; and GE writing and comm course: level 2, or GE foundation writing and info literacy course	
CSE 5912	Capstone Design: Game Design and Development	4	Must have credit for CSE 3541, or 5541; and 2501, or Philos 1338, or 2338; and CSE 3901, or 3902, or 3903; and GE writing and comm course: level 2, or GE foundation writing and info literacy course	Computer Graphics and Game Design (elective)
CSE 5913	Capstone Design: Computer Animation	4	Must have credit for CSE 3541, or 5541; and 2501, or Philos 1338, or 2338; and CSE 3901, or 3902, or 3903; and GE writing and comm course: level 2; or GE foundation writing and info literacy course	Computer Graphics and Game Design (elective)
CSE 5914	Capstone Design: Knowledge-Based Systems	4	Must have credit for CSE 3521, or 5521; and 2501, or Philos 1338, or 2338; and CSE 3901, or 3902, or 3903; and GE writing and comm	Artificial Intelligence (elective)

Course	Title	Credits	Prerequisites and Notes	Specialization Requirement
			course: level 2, or GE foundation writing and info literacy course	
CSE 5915	Capstone Design: Information Systems	4	Must have credit for CSE 3241, or 5241; and 2501, or Philos 1338, or 2338; and CSE 3901, or 3902, or 3903; and GE writing and comm course: level 2, or GE foundation writing and info literacy course	

Technical Electives and Specializations



Technical electives

All BS CIS students must choose a **specialization option**, which will determine a portion of a student's technical electives. Overall, each student must have 16 credit hours of technical elective coursework, of which 9 hours must be in CSE. Some of these 9 hours will be determined by the selected specialization. The remaining 7 hours of technical electives may be additional CSE course at the 3000-level or higher not already counting toward a degree requirement or select non-CSE courses at the 2000-level or higher.

Limitations for certain CSE courses

Students are limited in the number of hours they may take of CSE courses that are graded S/U or that focus on research.

Course	Title	Credit	Prerequisites and Notes
CSE 4251	The UNIX Programming Environment	1	Must have credit for CSE 2122, 2123, or 2231; and 2321; and enrollment in CSE, CIS, ECE, Engr Physics, or Data Analytics major, or CS minor. This course is graded S/U. No more than 2 of CSE 4251, 4252, 4253, and 4256 may be counted toward technical electives.
CSE 4252	Programming in C++	1	Must have credit for CSE 2123 or 2231; and 2321; and enrollment in CSE, CIS, ECE, Engr Physics, or Data Analytics major, or CS

Course	Title	Credit	Prerequisites and Notes
			minor. Not open to students with credit for 2122. This course is graded S/U. No more than 2 of CSE 4251, 4252, 4253, and 4256 may be counted toward technical electives. No more than 2 of CSE 4251, 4252, 4253, and 4256 may be counted toward technical electives.
CSE 4253	Programming in C#	1	Must have credit for CSE 2122, 2123, or 2231; and 2321; and enrollment in CSE, CIS, ECE, Engr Physics, or Data Analytics major, or CS minor. This course is graded S/U. No more than 2 of CSE 4251, 4252, 4253, and 4256 may be counted toward technical electives.
CSE 4256	Programming in Python	1	Must have credit for CSE 2122, 2123, or 2231; and 2321; and enrollment in CSE, CIS, ECE, Engr Physics, or Data Analytics major, or CS minor. This course is grades S/U. No more than 2 of CSE 4251, 4252, 4253, and 4256 may be counted toward technical electives.
CSE 4193	Individual Studies in Computer Science and Engineering	up to 2	Must have permission of instructor. This course is graded S/U. No more than 6 hours total of CSE 4193, 4998, and 4999 combined may be counted toward technical electives.
CSE 4998	Undergraduate Research in Computer Science and Engineering	up to 3	Permission of instructor. This course is graded S/U. No more than 6 hours total of CSE 4193, 4998, and 4999 combined may be counted toward technical electives.
CSE 4999	Computer Science and Engineering Research for Thesis	up to 6	Permission of instructor. This course is graded S/U. No more than 6 hours total of CSE 4193, 4998, and 4999 combined may be counted toward technical electives.

Non-CSE technical electives

If you are looking to use an outside department course as one of your technical electives, you are allowed to do so if you met one of the following qualifications:

- The course is a part of any completed minor.
- The course is listed on [the pre-approved list](#).
- You have successfully [petitioned](#) for an exception.

Non-Computer Science Core and Electives



In addition to the core CSE courses, BS CIS majors are expected to take the supplemental courses listed below. Unlike the BS CSE program, students are not required to take ECE 2020 (Introduction to Analog Systems and Circuits) nor MATH 2568 (Linear Algebra). These two courses are replaced with Math 1152 (Calculus II) and five credits worth of science electives. Note that Linear Algebra is still required for the AI specialization as it is a prerequisite for CSE 3521.

Non-Computer Science Core

Course	Title	Credits	Prerequisites and Notes
ECE 2060	Introduction to Digital Logic	3	Not open to students with credit for ECE 2000, 2000.02, 2000.07, 2001, 2010, or 2017
Math 1152	Calculus II		
Math 3345	Foundations of Higher Mathematics	3	Must have credit for CSE 2321 with a C- or higher; and have credit for 1152, 1161.xx, 1172, 1181H, 1534, 1544, or 4181H with a C- or higher.
Stat 3470	Introduction to Probability and Statistics for Engineers	3	Must have credit for Math 1152, 1161.xx, 1172, 1181H, or equiv, or permission of instructor. Not open to students with credit for 3440, 3450, 3450.01, 3450.02, or 3460.
Phys 1250	Mechanics, Work and Energy, Thermal Physic	5	Must have credit for or be enrolled in Math 1141, 1151, 1154, 1156, 1161, 1181H, or 4181H.
ASC 1100	Arts and Sciences Survey: General	1	Permission of department or instructor. This course is graded S/U.

Science Electives

BS CIS majors are given a large amount of freedom in choosing their five hours of science electives. Note that these electives are in addition to your Gen Ed science requirements, but many students choose classes from the same General Education list. To view the list of qualifying general education science courses:

1. Go to the [General Education Requirements](#) page
2. Select the new or legacy requirements
3. Select the attached Bachelor of Science document

Students can also browse courses in specific science departments by heading over to [Class Search](#).

General Education >

All students need 24 credits in General Education. For details of how these hours are broken down, please refer to the [General Education](#) page for engineers. OSU's [class search engine](#) also features Gen Ed filters. For instructions on how to use these filters, refer to [this instructional video](#).

>> Additional Resources

Sample Schedule >

Autumn	Spring
Year 1	
ArtsSci 1100 - 1 credit	CSE 2221 (need C- or better) - 4 credits
Math 1151 - 5 credits	Math 1152 - 5 credits
Physics 1250 - 5 credits	GE-Writing - 3 credits
CSE 1223 - 3 credits	Science - 5 credits
GE Launch Seminar - 1 credit	
Total of 15 credits	Total of 17 credits
Year 2	
CSE 2231 - 4 credits	CSE 2331 - 3 credits
CSE 2321 - 3 credits	CSE 2421 - 4 credits
Stat 3470 - 3 credits	Math 3345 - 3 credits
Foreign Language 1 - 4 credits	Foreign Language 2 - 4 credits
	GE-LVPA - 3 credits
Total of 14 credits	Total of 17 credits

Autumn	Spring
Year 3	
CSE 2431 - 3 credits	CSE 32X1 - 3 credits
CSE 390X - 4 credits	CSE 34X1 - 3 credits
ECE 2060 - 3 credits	CSE 35X1 - 3 credits
Foreign Language 3 - 4 credits	GE-Social Science - 3 credits
GE-History - 3 credits	CSE 2501 or Philos 2338 - 1 or 4 credits
	[GE-Theme if not Philos 2338 - 4 credits]
Total of 17 credits	Total of 16-17 credits
Year 4	
CSE 3341 - 3 credits	CSE 591X - 4 credits
Technical Elective - 3 credits	Technical Elective - 3 credits
Technical Elective - 3 credits	Technical Elective - 3 credits
Technical Elective - 3 credits	Technical Elective - 1 credit
GE -Diversity - 3 credits	GE-Theme - 4 credits
Total of 15 credits	Total of 15 credits

[Curriculum Sheet with Sample Schedule \(download\)](#)

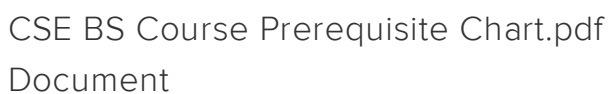


[Curriculum Sheet with Sample Schedule \(.docx\)](#)

[Prerequisite Visualization](#)



Capstone Pathway



A PDF printable version of the PNG image above.

Additionally, there is a **text-base description of this flowchart** designed with screen readers in mind.

An application to graduate must be submitted at the appropriate time prior to the graduation term. Filing the application in time will enable you to receive graduating-senior scheduling priority for your graduation term, while failure to

apply will prohibit you from graduating. To begin the graduation-application process, please contact your academic advisor one semester prior to your graduation term.

POCAT (Program Outcomes Achievement Test)

The Department of Computer Science and Engineering uses a comprehensive set of processes and tools to assess its programs to evaluate various aspects of the programs for the purposes of identifying and implementing improvements. To assist with this, during the graduation term, BS CSE, BS CIS, and BA CIS majors are required to complete the **exit test, POCAT**. Students will receive emails during their final semester about accessing and submitting the POCAT.

>> Department and Program Accreditations

ABET >

The Ohio State University is accredited by the **Higher Learning Commission of the North Central Association of Colleges and Schools**. The BS CIS program is not accredited by any commission of **ABET** and has not sought such accreditation.

Program Objectives and Outcomes >

Objectives:

- Graduates of the program will be employed in the computing profession, and will be engaged in learning, understanding, and applying new ideas and technologies as the field evolves.
- Graduates with an interest in, and aptitude for, advanced studies in computing will have completed, or be actively pursuing, graduate studies in computing.
- Graduates will be informed and involved members of their communities, and responsible engineering and computing professionals who take appropriate account, in their professional work, of such issues as privacy, security, copyright etc. in ways that are consistent with the ACM/IEEE Code of Conduct.

Outcomes: Students in the BS CIS program will attain:

- an ability to apply knowledge of computing, mathematics including discrete mathematics as well as probability and statistics;
- an ability to design and conduct experiments, as well as to analyze and interpret data;
- an ability to design, implement, and evaluate a software system, component, or process to meet desired needs within realistic constraints such as memory, runtime efficiency, as well as appropriate constraints related to economic, environmental, social, political, ethical, health and safety, and sustainability considerations;
- an ability to function on multi-disciplinary teams;
- an ability to identify, formulate, and solve computing problems;
- an understanding of professional, ethical, legal, security and social issues and responsibilities;
- an ability to communicate effectively with a range of audiences;
- an ability to analyze the local and global impact of computing on individuals, organizations, and society;
- a recognition of the need for, and an ability to engage in life-long learning and continuing professional development;
- a knowledge of contemporary issues;
- an ability to use the techniques, skills, and modern tools necessary for practice as a computing professional;
- an ability to analyze a problem, and identify and define the computing requirements appropriate to its solution;
- an ability to apply mathematical foundations, algorithmic principles, and computer science theory in the modeling and design of computer-based systems in a way that demonstrates comprehension of the trade-offs involved in design choices;
- an ability to apply design and development principles in the construction of software systems of varying complexity.



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Ohio State is in the process of revising websites and program materials to accurately reflect compliance with the law. While this work occurs, language referencing protected class status or other activities prohibited by Ohio Senate Bill 1 may still appear in some places. However, all programs and activities are being administered in compliance with federal and state law.

■ CONNECT

